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30 June 1981
SUPERSEDING
MIL-C-48456A(AR)
20 July 1976

MILITARY SPECIFICATION

COMPUTER UNIT, FIRE CONTROL: 11732400

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Computer Unit, Fire Control: 11732400 which is a unit of the M21 Ballistic Computer, 11732979.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical

MILITARY

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquistion, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

FSC: 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732400	Computer Unit, Fire Control
11743670	Motor - Generator

Inspection Equipment

11750202	Automatic Card and Unit Test Set
11750441	Adapter Computer Unit
11821520	Automatic Test Assembly - Central Processor Unit
11821748	Test Set - Immersion Leak, Purge and Fill
11821208	Fixture Chamber - Subsystem
11821457	Fixture Shock, Vertical and Horizontal
11821458	Shock Adapter, Plate Vib Horiz-Vert Plane Shock-Z
11823177	Unit Fixture-Computer, M60A3 Vib/Shock
11821471	Adapter - Handling Fixture, Altitude

Packaging Data Sheet

P11732400	Packaging of Computer Unit, Fire Control 11732400

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The fire control computer unit, herein refered to as the assembly, shall be manufactured in accordance with Drawing 11732400 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The assembly shall provide the following functions:

a. Ballistic computation
b. Lamp logic test of ammo select
c. Moving tank lamp logic
d. Servo drive control
e. Built in test
f. Output power and reference signals
g. ST BAR output

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	$73^{\circ} \pm 18^{\circ}F$
b.	Relative humidity	50 percent $\pm$ 30 percent
c.	Atmospheric pressure	28.5 + 2.0 $-$ 3.0 in. Hg.

3.2 First article. When specified (See 6.2), the contractor shall furnish sample units for first article inspection and approval (See 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

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TABLE I. Loads, power, and signals.

Item	Conditions	Characteristic 1/	Connections
1.0	Loads		Connected between:
1.1	Resistance	25 ohms ±10%, not less than 50 watts (W)	1J3-17 and 1J3-18
1.2	D Motor simulation circuit	see figure 1	see figure 1
1.3	MD Motor simulation circuit	see figure 2	see figure 2
1.4	Resistances (6)	300 ohms ±5% not less than 3W	Connected individually between 1J2-14, -15, -16, -17, -18, -19 and 1J2-24
1.5	Resistances (7)	620 ohms ±5% not less than 2W	Connected individually between 1J1-63, -11, -10, -9, -12, -8, -7, -14 and 1J1-60
1.6	Resistance	390 ohms ±5% not less than 1/2W	1J1-52 and 1J3-8
1.7	Resistance	390 ohms ±5% not less than 1/2W	1J1-54 and 1J3-8
1.8	Resistance	1K ohms ±5% not less than 1/2W	1J1-17 and 1J1-50
1.9	Wind Sensor simulation circuit	see figure 3	see figure 3
1.10	Cant Unit Resolver simulater circuit	see figure 4	see figure 4

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TABLE I. Loads, power, and signals. - Continued

Item	Conditions	Characteristic 1/	Connections
1.11	Resistance	75 ohms ±5% not less than 20W	Connected between 1J1-60 and return to 1J1-61
1.12	Servo Amplifier Output Conditioning circuits	see figure 5	see figure 5
1.13	VTACH Input circuit	see figure 6	see figure 6
1.14	115 V 400 Hz Signal Conditioning circuit	see figure 7	see figure 7
1.15	Feedback Conditioning circuits	see figure 8	see figure 8
1.16	Prime Power Conditioning circuit	see figure 9	see figure 9
2.0	Power		
2.1	17.5 vdc	Tolerance: +0.5, -0.0 vdc Ripple: 1V p-p max	see figure 9
2.2	24 vdc	Tolerance: ±2 vdc Ripple: 0.5 v p-p max	same as item 2.1
2.3	30 vdc	Tolerance: +0, -1 vdc Ripple: 2 v p-p max	same as item 2.1
3.0	Signals		Connected between:
3.1	BS (true) = +15 vdc in series with 500 ohm ±20% resistance BS (false) = -15 vdc in series with 10K ohm ±20% resistance	Tolerance: ±0.50 vdc	1J1-42 and return to 1J1-58

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TABLE I. Loads, power, and signals. - Continued

Item	Conditions	Characteristic 1/	Connections
3.2	$\overline{BS}$ (true) = +15 vdc in series with 500 ohm ±20% resistance $\overline{BS}$ (false) = -15 vdc in series with 10K ohm ±20% resistance	Tolerance: ±0.50 vdc	1J1-43 and return to 1J1-58
3.3	0 to 5.12 vrms 400 Hz (R_o Wear)	Tolerance: ±0.003 vrms, phase: 1	1J1-18 and return to 1J2-51
3.4	0 to 5.12 vrms 400 Hz (GCU Test, R_oP_1, R_oT_{A1})	Tolerance: ±0.003 vrms, phase: 1	1J1-21 and 1J1-66, 1J1-19 and return to 1J2-51
3.5	Jumper		1J1-28 to 1J1-24
3.6	AS6 (true) = 0.00 vdc in series with 20K ohm ±20% resistance (false) = -14.5 vdc in series with 200 ohm ±20% resistance	Tolerance: ±0.50 vdc	1J1-33 and return to 1J1-58
3.7	0.890 to 4.600 vrms 400 Hz (R_oAC)	Tolerance: ±0.002 vrms, phase: 1	1J1-44 and return to 1J3-23
3.8	+8.000 or -8.000 vdc (Manual Wind)	Tolerance: ±0.050 vdc Ripple: 0.02 v p-p max	1J1-45 and return to 1J1-59
3.9	-3.300 to +9.000 vdc (V_{TACH})	Tolerance: ±0.012 vdc Ripple: 0.01 v p-p max	see figure 6
3.10	-2.50 to +2.50 vdc (WDG1)	Tolerance: ±0.002 vdc	1J2-69 and return to 1J2-71

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TABLE I. Loads, power, and signals. - Continued

Item	Conditions	Characteristic 1/	Connections
3.11	BR (true) = 4.7 vdc in series with 6.2K ohm ±5% resistance BR (false) = 0.5 vdc	Tolerance: 0.5 ±0.2 vdc or 4.7 ±1.0 vdc Ripple: 0.02 v p-p max	1J3-55 and return to 1J3-52
3.12	LL2 (on) = 24 vdc in series with 1K ohm ±20% resistance LL2 (off) = open, no connection	Tolerance: ±1 vdc	1J2-66 and return to 1J1-58
3.13	STAB (on) = 24 vdc in series with 1K ohm ±20% resistance STAB (off) = open, no connection	Tolerance: ±1 vdc	1J2-72 and return to 1J1-58
3.14	0 to 5.0 vrms, 400 Hz	Tolerance: ±0.005 vrms Phase: 1 or 2 as required	see figure 8
3.15	0 to 5.0 vrms, 400 Hz	Tolerance: ±0.005 vrms Phase: 1 or 2 as required	see figure 8
3.16	0 to 5.0 vrms, 400 Hz	Tolerance: ±0.005 vrms Phase: 1 or 2 as required	see figure 8
3.17	+2.50 to -2.50 vdc (W_{DG2}, E_{DOT})	Tolerance: ±0.002 vdc	1J2-70, 1J3-20 and return to 1J2-71
3.18	0 to 5.12 vrms, 400 Hz (RoP_2, RoT_{A2})	Tolerance: ±0.003 vrms Phase: 1	1J1-20, -22 and return to 1J2-51
3.19	0 to 5.12 vrms, 400 Hz (D DOT, D DOT L)	Tolerance: ±0.010 vrms Phase: 1 or 2 as required	1J3-39, -66 and return to 1J2-51

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TABLE I. Loads, power, and signals. - Continued

Item	Conditions	Characteristic 1/	Connections
3.20	MT (on) = +15 vdc in series with 1K ohm ±20% resistance MT (off) = open, no connection	Tolerance: ±0.50 vdc	1J2-5 and return to 1J1-58
3.21	$\overline{MT}$ (on) = +15 vdc series with 1K ohm ±20% resistance $\overline{MT}$ (off) = open, no connection	Tolerance: ±0.50 vdc	1J2-6 and return to 1J1-58
3.22	0 to 5.120 vrms, 400 Hz (CRR1, CRR2)	Tolerance: ±0.003 vrms Phase: 1 or 2 as required	1J3-5, -11 and return to 1J1-59
3.23	BIT (on) = +13.5 vdc in series with 510 ohm ±20% resistance BIT (off) = -15 vdc in series with 10K ohm ±20% resistance	Tolerance: 13.5 ±0.5 vdc or -15.00 ±0.5 vdc	1J1-39 and return to 1J1-58
3.24	$\overline{BIT}$ (on) = +13.5 vdc in series with 510 ohm ±20% resistance $\overline{BIT}$ (off) = -15.00 vdc in series with 10K ohm ±20% resistance	Tolerance: +13.5 ±0.5 vdc or -15.00 ±0.5 vdc	1J1-62 and return to 1J1-58
3.25	0 to 5.12 vrms, 400 Hz ($-J_E$, J_D, $-Z_E$, Z_D)	Tolerance: ±0.004 vrms Phase: 1	1J1-15, -16, -46, -47 and return to 1J3-23

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TABLE I. Loads, power, and signals. - Continued

Item	Conditions	Characteristic 1/	Connections
3.26	True = 0.3 vdc for 150 ±50 milli-seconds (msec), thereafter open False = open (AS9, AS10, AS11, AS12)	Tolerance: ±0.3 vdc	1J2-9, -10, -11, or -12 and return to 1J2-13
3.27	MW (true) = +13.5 vdc in series with 500 ohm ±20% resistance MW (false) = -15.0 vdc in series with 10K ohm ±20% resistance	Tolerance: ±0.5 vdc	1J1-40 and return to 1J1-58
3.28	$\overline{MW}$ (true) = +13.5 vdc in series with 500 ohm ±20% resistance $\overline{MW}$ (false) = -15 vdc in series with 10K ohm ±20% resistance	Tolerance: ±0.5 vdc	1J1-41 and return to 1J1-58

1/ Phase of ac signals:
1 signals are in phase ±1° with proper output at 1J1-52 (see table II, item 1).
2 signals are 180° ±1° out of phase 1 signals.

3.3.1 Ballistic computation. (See 4.6.2.1)

3.3.1.1 Normal operation. With the loads (table I, item 1.0) applied, the power form (table I, item 2.2) applied, the signal sources (table I, items 3.14, 3.15 and 3.16) applied and with the following conditions satisfied,

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a. $V_{1J3-31} = -0.580 * V_{1J4-4} \pm 0.005$ vrms

b. $V_{1J3-37} = -1.464 * V_{1J4-5} \pm 0.007$ vrms

c. $V_{1J3-62} = -0.589 * V_{1J4-67} \pm 0.005$ vrms

the assembly shall provide the outputs of table IV after the input conditions of table III have been applied for not less than 1.0 second unless otherwise specified.

3.3.1.2 Battle range. With the loads, power form, signal sources and input conditions applied as specified in 3.3.1.1 for item 3 of table III, except the signal source of table I, item 3.11 (BR_L) is set to the True Condition, the voltage at 1J4-67 (D_L) shall be:

$V_{1J4-67} = -0.038 \pm 0.010$ vrms (at ambient temperature)

$= -0.038 \pm 0.021$ vrms (over full operating temperature range)

3.3.1.3 WDT input lockout. With the loads, power form, signal sources and input conditions applied as specified in 3.3.1.1 for item 29 of table III and the signal source of table I, item 3.9 (V_{TACH}) changed from 3.252 $\pm$ 0.012 to -9.000 $\pm$ 1.000 VDC after the signal source of table I, item 3.13 (Lead Lock 2) has been applied as specified in table III, note 10, the output at 1J4-5 (-D) shall change not greater than 15 mvrms for not less than 5.0 seconds.

3.3.1.4 Omega time of flight zero. With the loads, power form, signal sources and input conditions applied as specified in 3.3.1.1 for table III, item 20, the voltage at 1J4-69 (W - kwVw) TF shall be 0.000 $\pm$ 0.012 vrms. The measured voltage shall be defined as $WT_f(0)$.

3.3.2 Lamp logic test of ammo select. The assembly shall provide the outputs as specified in table V with the specified input conditions the power form of table I, item 2.2 and loads of table I, item 1 applied. (See 4.6.2.2)

3.3.3 Moving tank lamp logic. The assembly shall provide the outputs as specified in table VI with the specified input conditions the power form of table I, item 2.2 and loads of table I, item 1 applied. (See 4.6.2.3)

3.3.4 Servo drive control. (See 4.6.2.4)

3.3.4.1 Servo drive.

3.3.4.1.1 Positive drive. With the loads and power form applied as specified in 3.3.1.1 and the signal sources of table I, items 3.14, 3.15 and 3.16 applied with the following conditions satisfied,

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a. $V_{1J3-31} = -0.580 * V_{1J4-4} + 0.018 \pm 0.003$ Vrms

b. $V_{1J3-37} = -1.464 * V_{1J4-5} + 0.090 \pm 0.005$ Vrms

c. $V_{1J3-62} = -0.589 * V_{1J4-67} + 0.150 \pm 0.005$ Vrms

the assembly shall provide the following outputs after the input conditions of table III, item 3 have been applied for not less than 1.5 seconds:

d. (EMTR1 - EMTR2) V_{1J1-1} - 4.000 VDC

e. $-5 * \left(\frac{DMTR2 - DMTR1}{5}\right) \leq -1.41 * V_{1J1-1} + 3.000$ Vrms

f. $-5 * \left(\frac{MDM2 - MDM1}{5}\right) \leq -1.41 * V_{1J1-1} + 3.000$ Vrms

where:

g. The expression (EMTR1 - EMTR2) is the output measured across the load specified in figure 5.

h. The expression $\left(\frac{DMTR2 - DMTR1}{5}\right)$ is the output measured across the load specified in figure 5.

i. The expression $\left(\frac{MDM2 - MDM1}{5}\right)$ is the output measured across the load specified in figure 5.

3.3.4.1.2 Negative drive. With the loads and power form applied as specified in 3.3.1.1 and the signal sources of table I, items 3.14, 3.15 and 3.16 applied with the following conditions satisfied,

a. $V_{1J3-31} = -0.580 * V_{1J4-4} - 0.018 \pm 0.003$ Vrms

b. $V_{1J3-37} = -1.464 * V_{1J4-5} - 0.090 \pm 0.005$ Vrms

c. $V_{1J3-62} = -0.589 * V_{1J4-67} - 0.150 \pm 0.005$ Vrms

the assembly shall provide the following outputs after the input conditions of table III, item 3 have been applied for not less than 1.5 seconds:

d. (EMTR1 - EMTR2) $- V_{1J1-1}$ - 4.000 VDC

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e. $-5 * \left(\frac{DMTR2 - DMTR1}{5}\right) \leq 1.41 * V_{1J1-1} - 3.000$ Vrms

f. $-5 * \left(\frac{MDM2 - MDM1}{5}\right) \leq 1.41 * V_{1J1-1} - 3.000$ Vrms

3.3.4.2 Servo null. With the loads and power form applied as specified in 3.3.1.1 and the signal sources of table I, items 3.14, 3.15 and 3.16 applied with the following conditions satisfied,

a. $V_{1J3-31} = -0.580 * V_{1J4-4} \pm 0.003$ Vrms

b. $V_{1J3-37} = -1.464 * V_{1J4-5} \pm 0.005$ Vrms

c. $V_{1J3-62} = -0.589 * V_{1J4-67} \pm 0.005$ Vrms

the asembly shall meet the following conditions after the input conditions of table III, items 3, 7, 14 and 22 have been applied for not less than 1.0 second:

d. $\left|(EMTR1 - EMTR2) - 7.25 * V_{1J3-20}\right| \leq 1.5 + 1.21 * V_{1J4-4} + 0.6 * V_{1J3-20}$ VDC

e. $\left|-5 * \left(\frac{DMTR2 - DMTR1}{5}\right) + 28.9 * V_{1J3-39}\right| \leq 1.0 - 0.87 * V_{1J4-5} + 5.7 * V_{1J3-39}$ vrms

f. $\left|-5 * \left(\frac{MDM2 - MDM1}{5}\right) - 10 * V_{1J3-66}\right| \leq 0.06 + 0.18 * V_{1J4-67} + 1.1 * V_{1J3-66}$ vrms

3.3.5 Built in test. With the input signals applied as specified for each item of table VII, the input power of table I, item 2 applied and the output loads of table I, item 1 applied, the output conditions of table VII shall be met for any item order after all of the inputs have been applied for not less than 2.0 seconds. (See 4.6.2.5)

3.3.6 Output power and reference signals. With the loads of table I, item 1 and the power form of table I, item 2.2 applied, the output power and reference signals shall be as specified in table II. (See 4.6.2.6)

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3.3.6.1 Output power and reference signal check. With input power of table I, item 2.2 applied and the output power and reference signals as specified in table II, the voltage (W.S. Fail) at 1J1-14 shall be the voltage of table II, item 6 ±1.00 VDC. The voltage (R.P. Pwr Fail) at 1J4-57 shall be 0.3 ± 0.3 VDC. The output voltages at 1J1-14 and 1J4-57 shall be referenced to 1J2-51.

3.3.7 ST BAR output. With the power, table I, item 2 and the signals table I, items 3.23 (OFF) and 3.24 (ON) applied, the output at 1J3-60 shall be 4.5 ±1.0 vdc. With the power, table I, item 2 and the signals table I, items 3.23 (ON) and 3.24 (OFF) applied, the output shall be 0.4 ±0.4 vdc at 1J3-60. The output at 1J3-60 shall be referenced to 1J3-26. (See 4.6.2.7)

TABLE II. Output power and reference signals.

Item	Output	Characteristic 1/
1	1J1-52 and return to 1J1-59	Sine wave, 5.000 ±0.007 (0.030) vrms, 400 ±1 Hz 2/
2	1J1-54 and return to 1J1-59	Sine wave, item 1 ±0.006 (0.015) vrms, 180° out of phase with item 1 ±1.0°
3	1J3-33 and return to 1J3-32	Sine wave, 115.0 ±10.0 (15.0) vrms, 180° out of phase with item 1 ±10°
4	1J1-70 and return to 1J2-13	5.2 ± 0.5 (0.7) vdc
5	1J3-78 and return to 1J1-68	12.0 ± 0.5 (0.6) vdc
6	1J2-24 and return to 1J2-25	23.00 ± 2.00 (3.00) vdc
7	1J2-7 and return to 1J1-50	15.000 ± 0.150 (0.320) vdc
8	1J1-56 and return to 1J1-50	-15.000 ± 0.150 (0.320) vdc
9	1J1-64 and return to 1J2-46	30.0 ± 3.0 (4.5) vdc

1/ Numbers in parentheses are tolerances at temperature extremes of 3.5.3.
2/ 398 ± 2.1 Hz at -25°F
402 ± 2 Hz at 125°F

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TABLE III. Ballistic computation inputs. 1/

Item	13/ R_o(AC)IN 1J1-44	6/ AS6 1J1-33	5/ APDS 1J2-9	5/ HEAT 1J2-10	5/ HEP 1J2-11	5/ FSDS 1J2-12	8/ STAB 1J2-78	9/ MT
1	1.277	1	1	0	0	0	0	1
2	3.032	1	1	0	0	0	0	1
3	1.277	1	1	0	0	0	0	1
4	1.968	1	1	0	0	0	0	1
5	1.968	1	1	0	0	0	0	1
6	3.032	1	1	0	0	0	0	1
7	4.574	1	1	0	0	0	0	1
8	3.723	0	1	0	0	0	0	1
9	1.277	1	0	1	0	0	0	1
10	3.032	1	0	1	0	0	0	1
11	1.277	1	0	0	1	0	0	1
12	1.968	1	0	0	1	0	0	1
13	3.032	1	0	0	1	0	0	1
14	3.298	1	0	0	1	0	0	1
15	1.277	1	0	0	0	1	0	1
16	1.968	1	0	0	0	1	0	1
17	1.968	1	0	0	0	1	0	1

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TABLE III. Ballistic computation inputs. 1/ - Continued

Item	13/ R_o(AC)IN 1J1-44	6/ AS6 1J1-33	5/ APDS 1J2-9	5/ HEAT 1J2-10	5/ HEP 1J2-11	5/ FSDS 1J2-12	8/ STAB 1J2-78	9/ MT
18	3.032	1	0	0	0	1	0	1
19	3.032	1	0	1	0	0	0	1
20	1.968	1	0	1	0	0	0	1
21	1.968	1	0	1	0	0	0	1
22	1.968	1	0	1	0	0	0	1
23	1.968	1	0	1	0	0	0	1
24	1.968	1	0	1	0	0	1	1
25	1.968	1	0	1	0	0	1	1
26	1.968	1	0	1	0	0	1	1
27	1.968	1	0	1	0	0	0	1
28	1.277	1	1	0	0	0	0	1
29	1.277	1	0	0	1	0	0	1
30	1.277	1	0	0	0	1	0	1
31	1.968	1	0	0	1	0	0	1
32	1.968	1	0	1	0	0	0	1
33	1.968	1	0	1	0	0	0	0

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TABLE III. Ballistic computation inputs. 1/ - Continued

Item	10/ LL 1J2-66	11/ V_{TACH}	12/ WDG1 1J2-69	14/ W_{DG2} 1J2-70 E DOT 1J3-20	2/ GCU TEST 1J1-66 R_oP_1 1J1-21 R_oT_{A1} 1J1-19	4/ R_oP_2 1J1-22 R_oT_{A2} 1J1-20	15/ CRR1 1J3-5 CRR2 1J3-11
1	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0
10	0	9.000	0	0	0	0	0
11	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0

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TABLE III. Ballistic computation inputs. 1/ - Continued

Item	10/ LL 1J2-66	11/ V_{TACH}	12/ WDG1 1J2-69	14/ W_{DG2} 1J2-70 E DOT 1J3-20	2/ GCU TEST 1J1-66 R_oP_1 1J1-21 R_oT_{A1} 1J1-19	4/ R_oP_2 1J1-22 R_oT_{A2} 1J1-20	15/ CRR1 1J3-5 CRR2 1J3-11
18	0	0	0	0	0	0	0
19	0	0	0	0	0	0	0
20	0	0	0	0	0	0	0
21	1	0	0	0	0	0	0
22	1	-1.084	2.105	0	0	0	0
23	1	-3.252	2.105	0	0	0	0
24	1	-3.252	2.105	1.000	0	0	0
25	1	-3.252	-2.105	-1.000	0	0	0
26	1	0	0	0	0	0	0
27	1	1.084	0	0	0	0	0
28	1	4.336	0	0	0	0	0
29	1	3.252	0	0	0	0	0
30	1	4.336	0	0	0	0	0
31	0	0	0	0	3.000	2.000	0
32	0	0	0	0	5.000	5.000	3.400
33	0	0	0	0	0	0	3.400

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TABLE III. Ballistic computation inputs. 1/ - Continued

Item	16/ D DOT IN 1J3-39 D_LDOT 1J3-66	3/ R_oWear 1J1-18	7/ Manual Wind 1J1-45	18/ Bore Sight	19/ Manual Wind Select	17/ Zeroing 1J1-15, 16	17/ Common Zeroing 1J1-46, 47
1	0	0	0	1	1	0	0
2	0	0	0	1	1	5.000	5.000
3	0	0	0	0	1	0	0
4	0	0	0	0	1	0	0
5	0	5.000	8.000	0	1	1.000	1.000
6	0	0	0	0	1	0	0
7	0	0	0	0	1	0	0
8	0	0	0	0	1	0	0
9	0	1.000	20/	0	0	0	0
10	0	0	0	0	1	0	0
11	0	0	0	0	1	0	0
12	0	0	8.000	0	1	0	0
13	0	0	0	0	1	0	0
14	0	0	0	0	1	0	0
15	0	0	0	0	1	0	0
16	0	0	-8.000	0	1	0	0
17	0	5.000	0	0	1	0	0

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TABLE III. Ballistic computation inputs. 1/ - Continued

Item	16/ D DOT IN 1J3-39 D_LDOT 1J3-66	3/ R_oWear 1J1-18	7/ Manual Wind 1J1-45	18/ Bore Sight	19/ Manual Wind Select	17/ Zeroing 1J1-15, 16	17/ Common Zeroing 1J1-46, 47
18	0	0	0	0	1	0	0
19	0	0	8.000	0	1	0	0
20	0	0	0	0	1	0	0
21	0	0	0	0	1	0	0
22	0	0	0	0	1	0	0
23	0	0	0	0	1	0	0
24	0.300	0	0	0	1	0	0
25	-0.300	0	0	0	1	0	0
26	0	0	0	0	1	0	0
27	0	0	0	0	1	0	0
28	0	0	0	0	1	0	0
29	0	0	0	0	1	0	0
30	0	0	0	0	1	0	0
31	0	2.000	0	0	1	0	0
32	0	0	0	0	1	0	0
33	0	0	0	0	1	0	0

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TABLE III. Ballistic computation inputs. 1/ - Continued

1/ The following inputs are constant for all items.

a. Table I, Item 3.23, OFF Condition
b. Table I, Item 3.24, ON Condition
c. Table I, Item 3.5
d. Table I, Item 3.11, False Condition

2/ Table I, Item 3.4

3/ Table I, Item 3.3

4/ Table I, Item 3.18

5/ 1 = Table I, Item 3.26, True Condition
0 = Table I, Item 3.26, False Condition

6/ 1 = Table I, Item 3.6, True Condition
0 = Table I, Item 3.6, False Condition

7/ Table I, Item 3.8

8/ 1 = Table I, Item 3.13, ON Condition
0 = Table I, Item 3.13, OFF Condition

9/ 1 = Table I, Item 3.20, ON Condition
0 = Table I, Item 3.21, OFF Condition

1 = Table I, Item 3.20, OFF Condition
0 = Table I, Item 3.21, ON Condition

10/ 1 = Table I, Item 3.12, ON Condition. Apply all inputs for at least 10 seconds with LL2 in OFF condition. Then apply LL2 (ON). The measurement of 1J4-5 must be taken at 1.0 ± 0.3 seconds after applying LL2 (ON). The output at 1J4-5 for Item 23 shall not change more than 30 mvrms from the initial measured value 15 ± 1 seconds after LL2 (ON) is applied.
0 = Table I, Item 3.12, OFF Condition

11/ Table I, Item 3.9 (Measure voltage V_{TACH} between source and 15K 0.05% resistor, not at 1J2-34.) (see figure 6).

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TABLE III. Ballistic computation inputs. 1/ - Continued

12/ Table I, Item 3.10

13/ Table I, Item 3.7

14/ Table I, Item 3.17

15/ Table I, Item 3.22

16/ Table I, Item 3.19

17/ Table I, Item 3.25

18/ 1 = Table I, Item 3.1, True Condition: Table I. Item 3.2, False Condition.
0 = Table I, Item 3.1, False Condition: Table I, Item 3.2, True Condition.

19/ 1 = Table I, Item 3.27, True Condition
Table I, Item 3.28, False Condition

0 = Table I, Item 3.27, False Condition
Table I, Item 3.28, True Condition

20/ Auto Wind problem. Allow at least 45 seconds after application of input power, Table I, Item 2.2, (for voltages to settle) before verifying outputs.

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TABLE IV. Ballistic computation outputs.

Item	E-VRMS 1J4-4	TOL (Ambient)	TOL (Temp)
1	-0.001	0.004	0.005
2	-0.028	0.004	0.005
3	0.288	0.008	0.014
4	0.448	0.009	0.015
5	-0.523	0.010	0.017
6	0.757	0.011	0.022
7	1.350	0.017	0.037
8	1.069	0.014	0.030
9	0.539	0.009	0.016
10	1.857	0.011	0.024
11	1.481	0.016	0.023
12	2.655	0.025	0.036
13	5.251	0.060	0.085
14	6.078	0.077	0.110
15	0.276	0.008	0.014
16	0.419	0.009	0.015
17	0.445	0.009	0.015
18	0.670	0.011	0.022
19	1.857	0.011	0.024
20	0.935	0.009	0.018
21	0.935	0.009	0.018
22	0.935	0.009	0.018
23	0.935	0.009	0.018
24	0.935	0.009	0.018
25	0.935	0.009	0.018
26	0.935	0.009	0.018
27	0.935	0.009	0.018
28	0.288	0.008	0.014
29	1.481	0.016	0.023
30	0.276	0.008	0.014
31	2.535	0.027	0.039
32	0.935	0.012	0.020
33	-3.416	0.020	0.027

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TABLE IV. Ballistic computation outputs. - Continued

Item	D-VRMS 1J4-5	TOL (Ambient)	TOL (Temp)	DL-VRMS 1J4-67	TOL (Ambient)	TOL (Temp)
1	-0.003	0.004	0.005	0.005	0.004	0.005
2	0.024	0.004	0.005			
3	0.008	0.009	0.020			
4	0.030	0.009	0.020			
5	-1.054	0.016	0.029			
6	0.051	0.009	0.020			
7	0.077	0.009	0.017			
8	0.064	0.009	0.017			
9	-0.137	0.026	0.046			
10	0.024	0.010	0.021			
11	0.052	0.009	0.018			
12	-0.805	0.015	0.040	0.789	0.016	0.041
13	0.215	0.014	0.022			
14	0.246	0.016	0.024			
15	-0.002	0.009	0.020			
16	0.065	0.012	0.025			
17	0.014	0.009	0.020			
18	0.024	0.009	0.020			
19	-0.831	0.016	0.037			
20	0.014	0.009	0.019			
21	0.014	0.011	0.021			
22	-1.006	0.016	0.037			
23	-3.050	0.039	0.070			
24	-3.050	0.062	0.098			
25	3.078	0.062	0.098			
26	0.014	0.036	0.050			
27	1.034	0.016	0.039			
28	1.686	0.023	0.056			
29	3.084	0.054	0.088			
30	1.64	0.023	0.056			
31	0.106	0.009	0.017			
32	0.014	0.009	0.020			
33	-3.381	0.022	0.029	3.365	0.023	0.030

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TABLE V. Lamp logic tests of ammo select.

Item	Input Conditions	Outputs (vdc) 1/			
		1J2-14	1J2-15	1J2-16	1J2-17
1	Table III, Item 1	+0.3 ± 0.28	V2 ± 0.5	V2 ± 0.5	V2 ± 0.5
2	Table III, Item 9	V2 ± 0.5	+0.3 ± 0.28	V2 ± 0.5	V2 ± 0.5
3	Table III, Item 11	V2 ± 0.5	V2 ± 0.5	+0.3 ± 0.28	V2 ± 0.5
4	Table III, Item 15	V2 ± 0.5	V2 ± 0.5	V2 ± 0.5	+0.3 ± 0.28

1/ V2 = Table II, Item 6

TABLE VI. Moving tank lamp logic.

Item	Input Conditions		Outputs (vdc) 1/	
	1J2-5	1J2-6	1J2-18	1J2-19
1	Table I, Item 3.20 MT (on)	Table I, Item 3.21 $\overline{MT}$ (off)	0.30 ± 0.28	V2 ± 0.5
2	Table I, Item 3.20 MT (off)	Table I, Item 3.21 $\overline{MT}$ (on)	V2 ± 0.5	0.30 ± 0.28

1/ V2 = Table II, Item 6

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TABLE VII. Built-in test.

	Inputs 1/						
Item	2/ R_O(AC)IN 1J1-44	3/ AS6 1J1-33	4/ APDS 1J2-9	4/ HEAT 1J2-10	4/ HEP 1J2-11	4/ FSDS 1J2-12	5/ E DOT IN 1J3-20 W_{DG2} 1J2-70
1	1.9415	1	1	0	0	0	0
2	1.9415	0	1	0	0	0	0
3	1.9415	1	0	1	0	0	0
4	1.9415	1	0	0	1	0	0
5	1.9415	1	0	0	0	1	0
6	1.9415	1	0	1	0	0	3.000
7	1.9415	1	0	1	0	0	0
8	1.9415	1	0	1	0	0	0
9	1.975	1	0	1	0	0	0
10	1.817	1	0	1	0	0	0
11	1.9415	1	0	1	0	0	0
12	1.9415	1	0	1	0	0	0
13	1.9415	1	0	1	0	0	0
14	1.9415	1	0	1	0	0	0

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TABLE VII. Built-in test. - Continued

	Inputs 1/					
Item	6/ D DOT IN 1J3-39 D_LDOT 1J3-66	7/ GCU TEST 1J1-66 R_oP_1 1J1-21 R_oT_{A1} 1J1-19	8/ R_oP_2 1J1-22 R_oT_{A2} 1J1-20	DFDBK 1J3-37	EFDBK 1J3-31	11/ CRR1 1J3-5 CRR2 1J3-11
1	0	0	0	9/	15/	-1.843
2	0	0	0	9/	15/	-1.875
3	0	0	0	9/	15/	-2.378
4	0	0	0	9/	15/	-2.512
5	0	0	0	9/	15/	-1.826
6	0	0	0	9/	15/	-2.378
7	0.450	-0.040	-0.040	9/	15/	-2.378
8	0	-0.120	-0.120	9/	15/	-2.378
9	0	0	0	9/	15/	-2.378
10	0	0	0	9/	15/	-2.378
11	0	0	0	9/	16/	-2.378
12	0.450	0	0	10/	15/	-2.378
13	0	0	0	9/	15/	-2.080
14	0	0	0	9/	15/	-2.573

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TABLE VII. Built-in test. - Continued

Item	Outputs				
	12/ O.U. Fail 1J1-11	12/ R.P. Fail 1J1-10	12/ R.F. Fail 1J1-9	12/ C.U. Fail 1J1-12	13/ Comp. Fail 1J1-8
1	1	1	1	1	1
2	1	1	1	1	1
3	1	1	1	1	1
4	1	1	1	1	1
5	1	1	1	1	1
6	1	1	1	1	0
7	1	1	1	1	0
8	1	1	1	1	1
9	1	1	0	1	1
10	1	1	0	1	1
11	0	1	1	1	1
12	1	0	1	1	1
13	1	1	1	0	1
14	1	1	1	1	0

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TABLE VII. Built-in test. - Continued

	Outputs				
Item	12/ W.S. Fail 1J1-14	12/ GCU Fail 1J3-63	12/ Sys OK 1J1-7	14/ D mVrms 1J4-5	14/ E mVrms 1J4-4
1	1	1	0	434 ±13 (32)	426 ±7 (10)
2	1	1	0	464 ±13 (33)	456 ±8 (11)
3	1	1	0	924 ±17 (46)	917 ±11 (16)
4	1	1	0	2319 ±36 (94)	2315 ±34 (40)
5	1	1	0	408 ±13 (62)	399 ± 7 (30)
6	1	1	1	N/A	N/A
7	1	1	1	N/A	N/A
8	1	0	1	N/A	N/A
9	1	1	1	N/A	N/A
10	1	1	1	N/A	N/A
11	1	1	1	N/A	N/A
12	1	1	1	N/A	N/A
13	1	1	1	N/A	N/A
14	1	1	1	N/A	N/A

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TABLE VII. Built-in test. - Continued

1/ The following inputs are constant for all items:
a. Table I, Item 3.23, BIT (ON)

b. Table I, Item 3.24, $\overline{\text{BIT}}$ (OFF)
c. Table I, Item 3.20, MT (ON)

d. Table I, Item 3.21, $\overline{\text{MT}}$ (OFF)
e. Table I, Item 3.12, LL2 (OFF)
f. Table I, Item 3.13, STAB (OFF)
g. Table I, Item 3.27, MW (TRUE)

h. Table I, Item 3.28, $\overline{\text{MW}}$ (FALSE)
i. Table I, Item 3.1, BS (FALSE)

j. Table I, Item 3.2, $\overline{\text{BS}}$ (TRUE)
k. Table I, Item 3.8 = -8.000 ± 0.050 VDC
l. Table I, Item 3.25 = 5.000 ± 0.004 VRMS, phase 1
m. Table I, Item 3.3 = 5.000 ± 0.005 VRMS, phase 1
n. Table I, Item 3.11 BR (FALSE)
o. Table I, Item 3.16 applied such that

$$V_{1J3-62} = -0.589 * V_{1J4-67} \pm 0.005 \text{ VRMS}$$

p. Table I, Item 3.9 (VTACH) = -3.000 ± 0.012 VDC
q. Table I, Item 3.10 (WDG1) = +2.50 ± 0.010 VDC
r. Table I, Item 3.5

2/ Table I, Item 3.7
3/ 1 = Table I, Item 3.6 True Condition
0 = Table I, Item 3.6 False Condition
4/ 1 = Table I, Item 3.26 True Condition
0 = Table I, Item 3.26 False Condition
5/ Table I, Item 3.17
6/ Table I, Item 3.19
7/ Table I, Item 3.4
8/ Table I, Item 3.18
9/ Table I, Item 3.15 applied such that $V_{1J3-37} = -1.464 * V_{1J4-5} \pm$ 0.005 VRMS
10/ Table I, Item 3.15 applied such that $V_{1J3-37} = -1.645 * V_{1J4-5} \pm$ 0.005 VRMS
11/ Table I, Item 3.22
12/ 1 = Table II, Item 6 ± 1.00 VDC, with return to 1J3-26.
0 = +0.5 ± 0.5 VDC, with return to 1J3-26.
13/ 1 = (Table II, Item 6 - 0.65) ± 1.00 VDC, with return to 1J3-26.
0 = +0.5 ± 0.5 VDC, with return to 1J3-26.
14/ Values in parentheses are tolerances at temperature extremes of 3.5.3.
15/ Table I, Item 3.14 applied such that $V_{1J3-31} = 0.580 * V_{1J4-4} \pm$ 0.003 VRMS
16/ Table I, Item 3.14 applied such that $V_{1J3-31} = -0.449 \pm 0.003$ VRMS

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3.4 Sealing. With an internal nitrogen pressure of 5.0 ± 0.1 psig, the assembly shall exhibit a pressure drop not greater than 0.2 psig for a period of not less than 2 hours. The pressurized assembly shall be purged with dry nitrogen specified as BB-N-411, type I, grade B, class 1 with a dewpoint not higher than -25°F. Final internal pressure of nitrogen shall be 0.0 + 2.2 - 0.0 psig. (See 4.6.4)

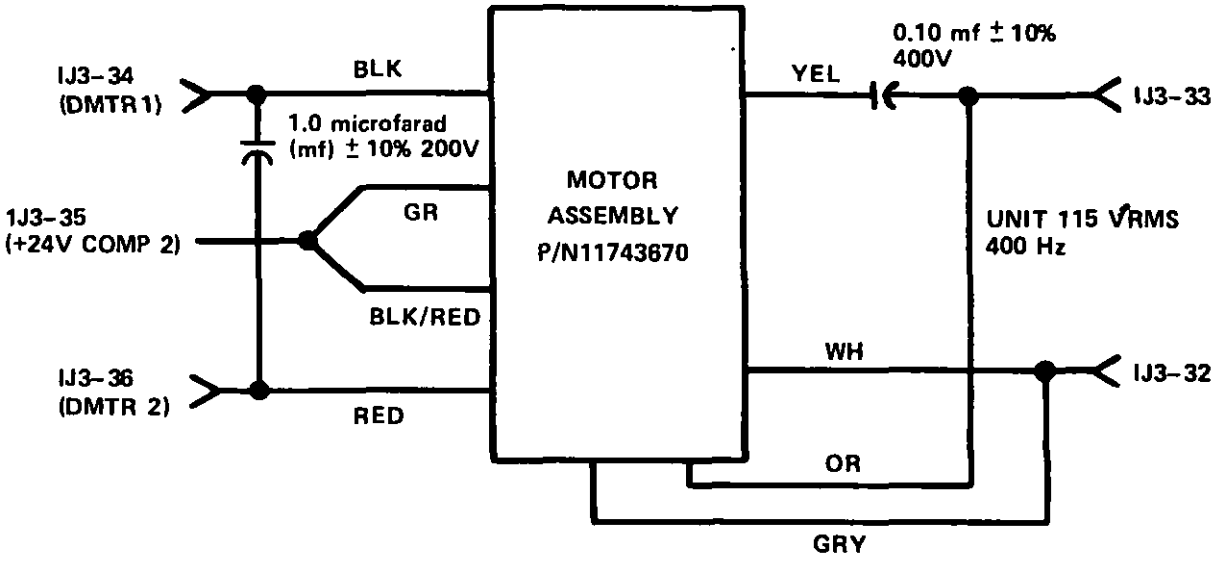


FIGURE 1. D motor load circuit.

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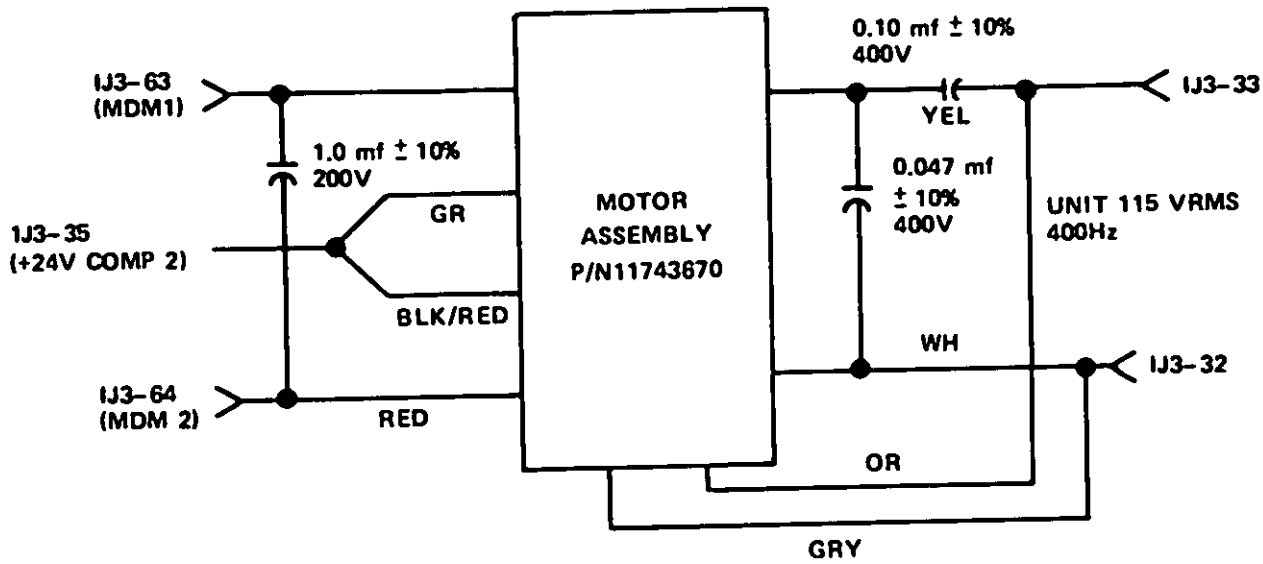


FIGURE 2. MD motor load circuit.

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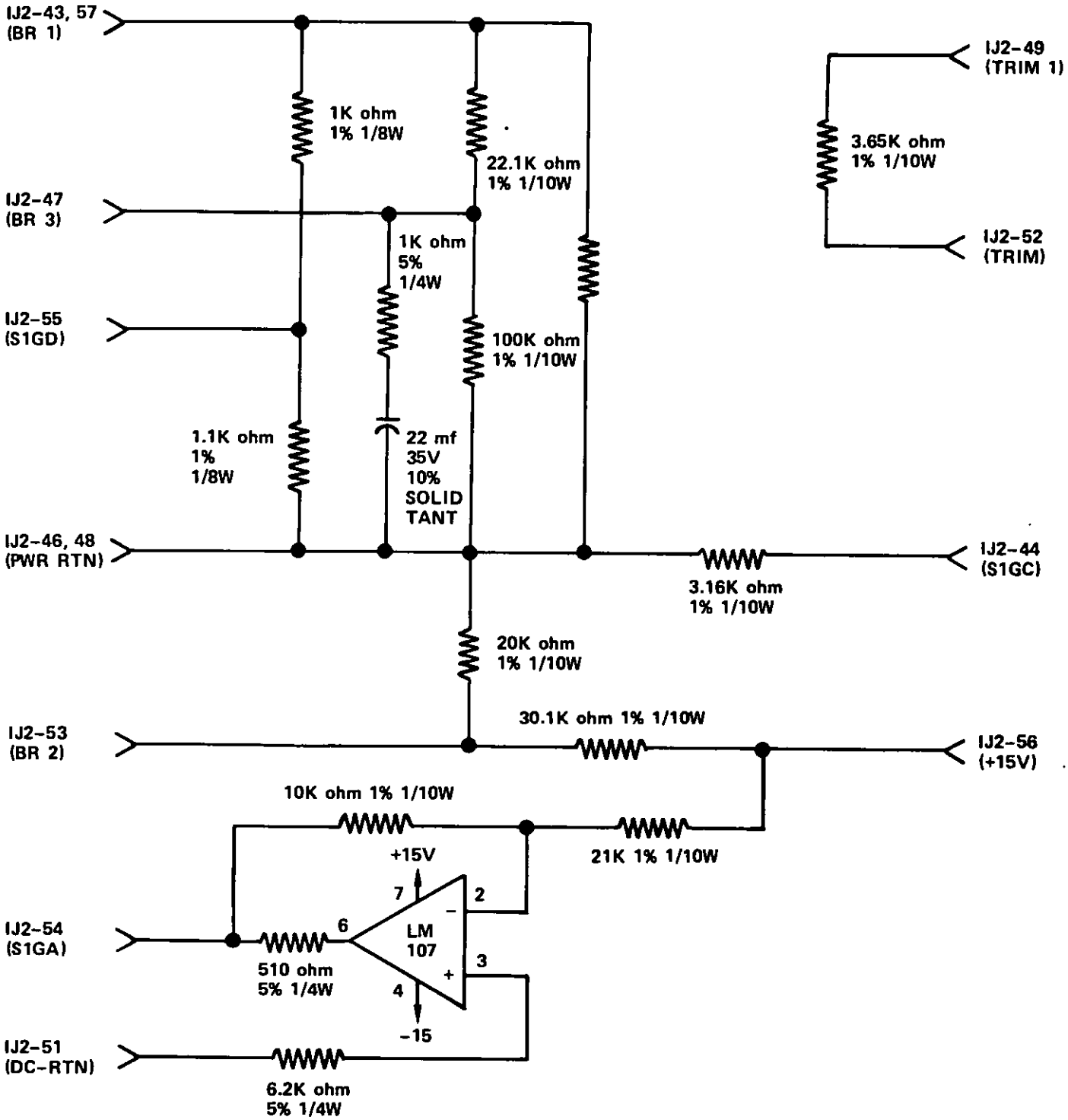


FIGURE 3. Wind sensor simulator circuit.

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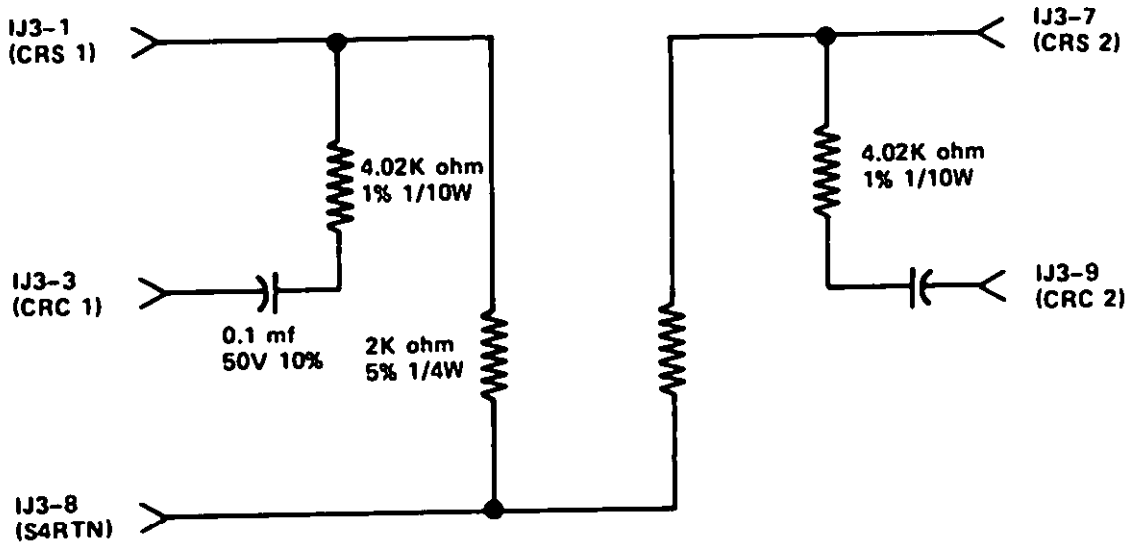


FIGURE 4. Cant unit resolver simulator circuit.

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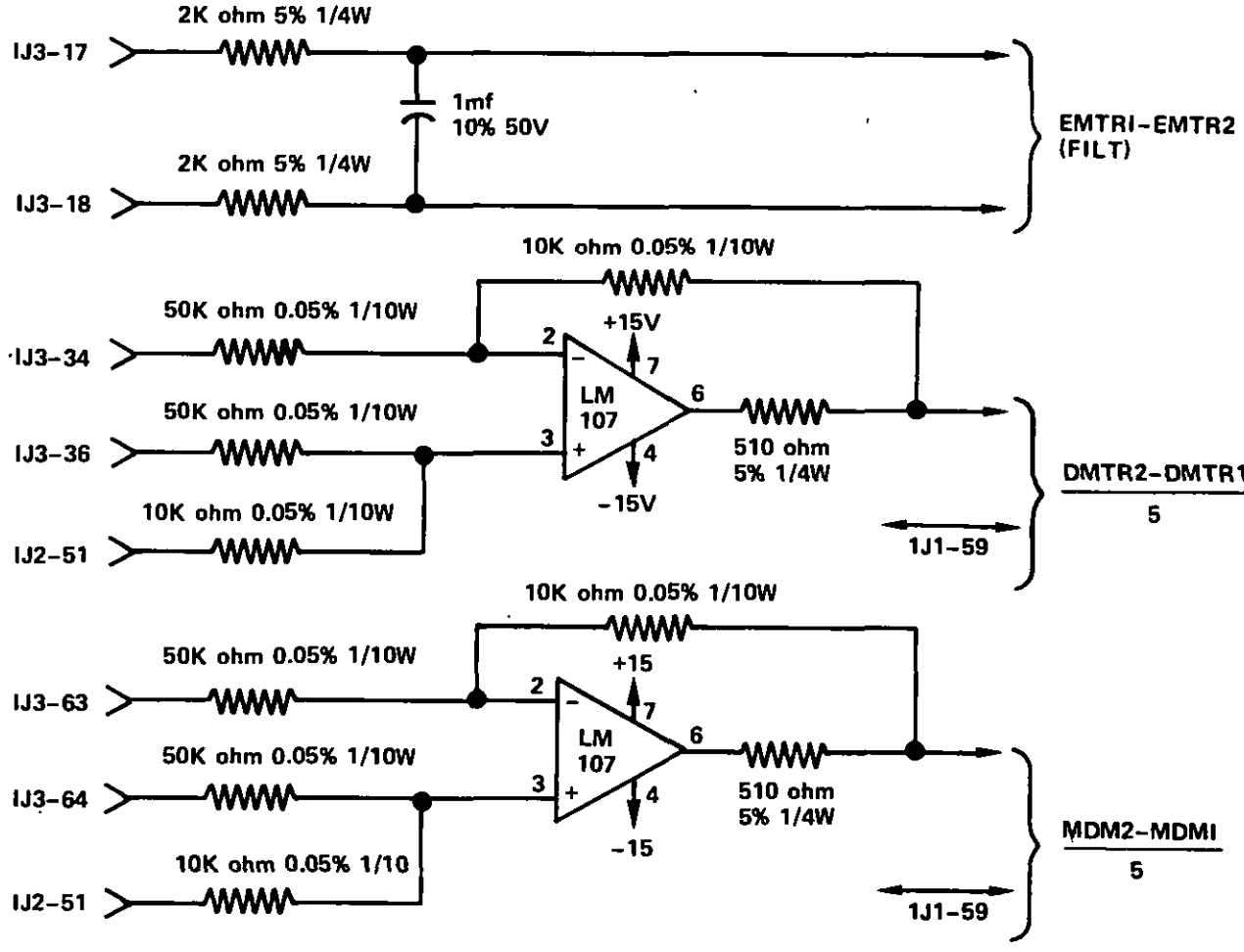


FIGURE 5. Servo amplifier output conditioning circuits.

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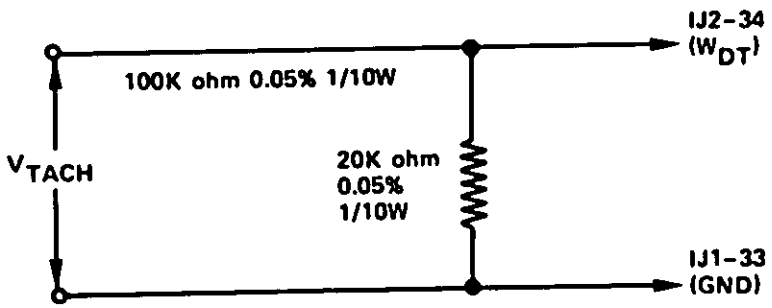


FIGURE 6. VTACH input circuit.

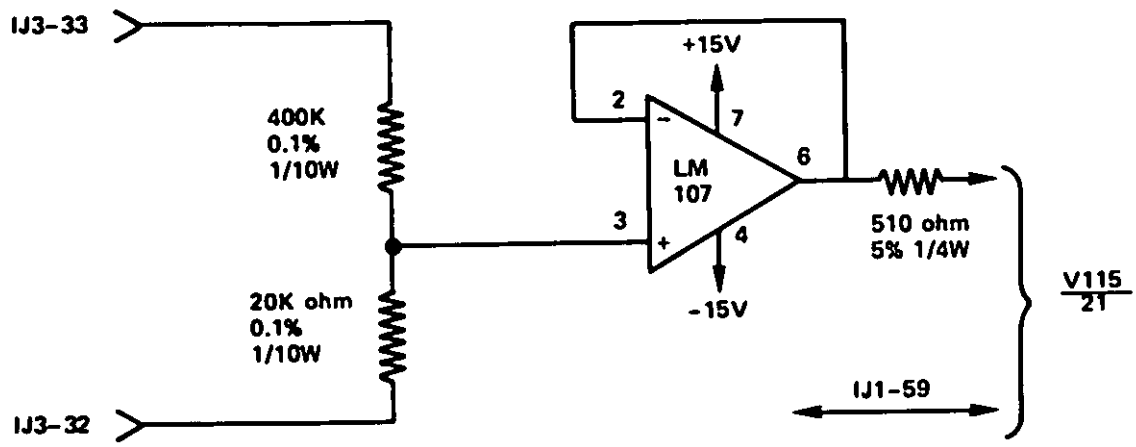


FIGURE 7. 115V 400 Hz signal conditioning circuit.

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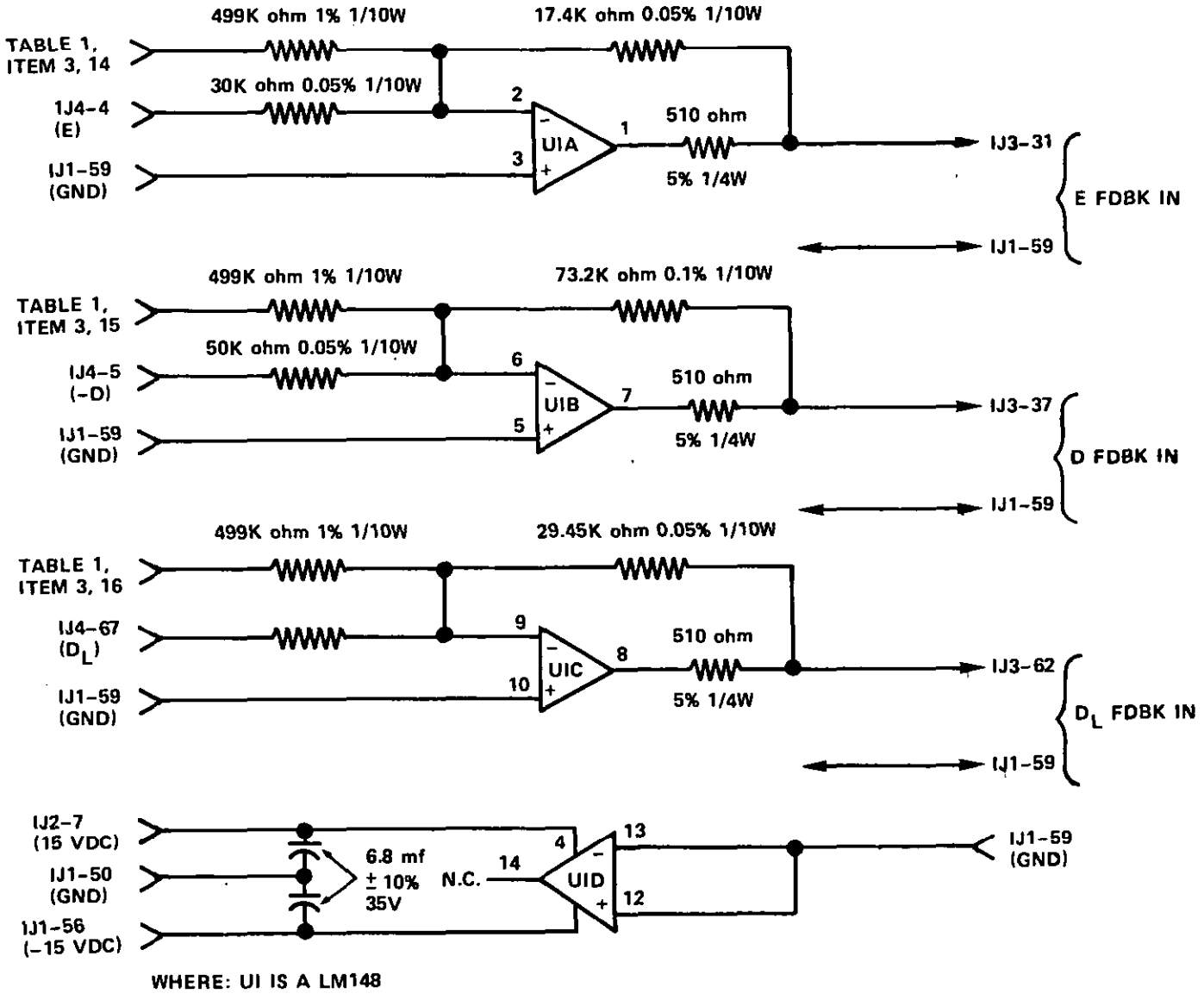
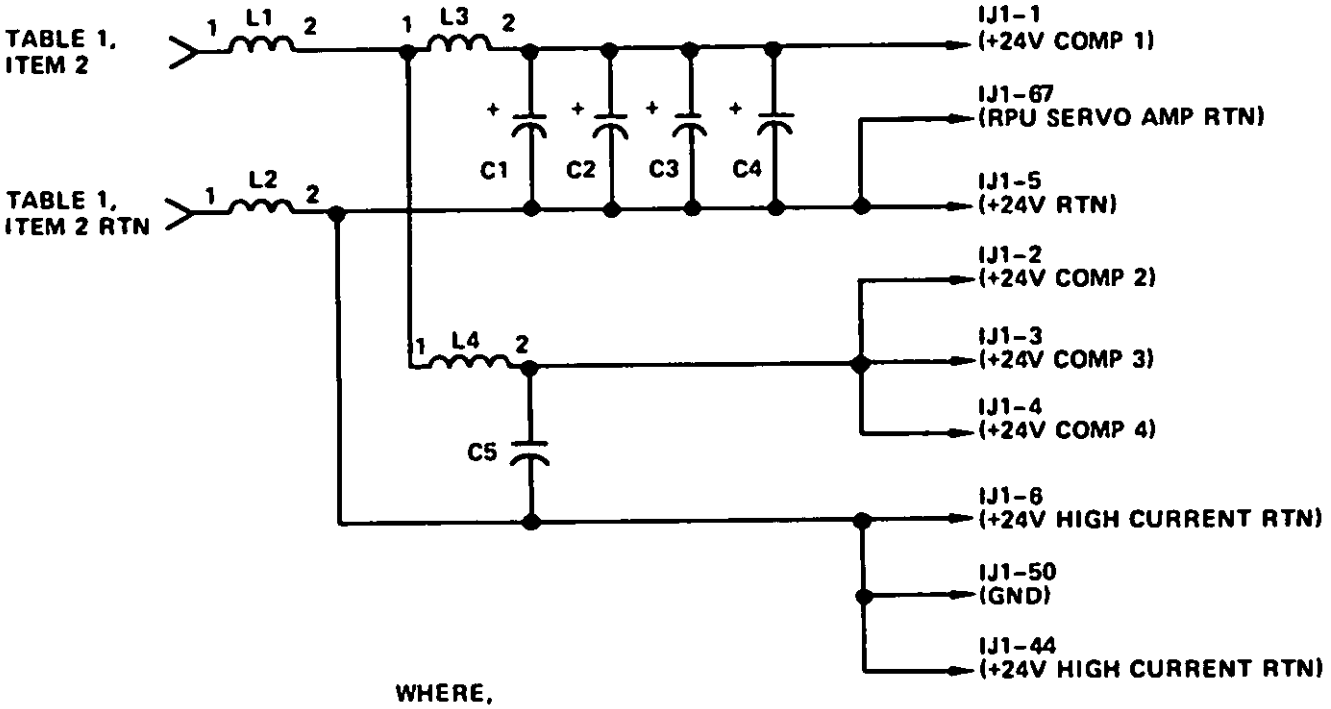


FIGURE 8. Feedback conditioning circuits.

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WHERE,

L1, L2, L3, L4 = 100 microhenry (mh) ± 10% 5 AMP (11732391)
C1, C2, C3, C4, C5 = 18 mf ± 10% 125V (M39006/09-5030)

FIGURE 9. Prime power conditioning.

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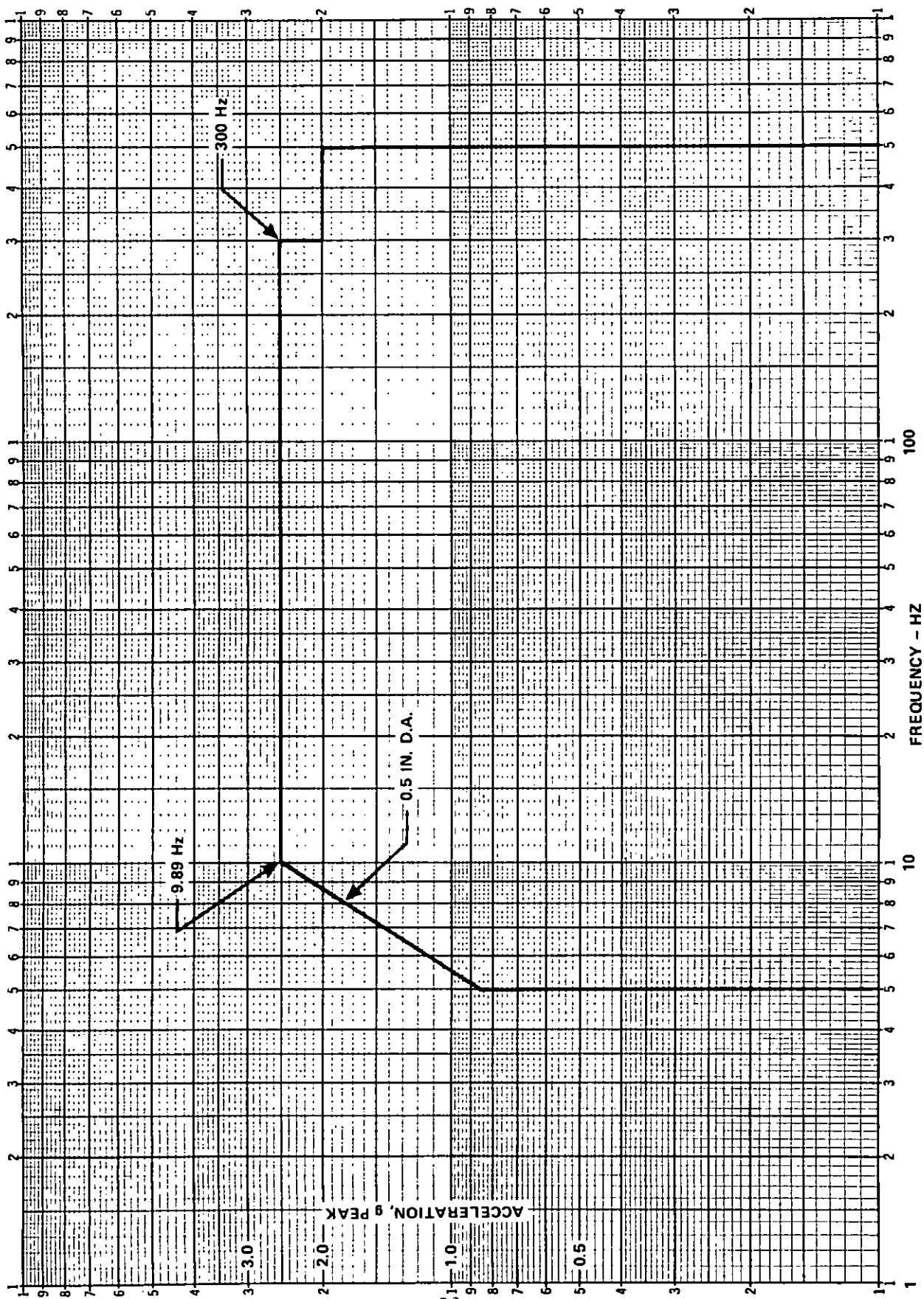


FIGURE 10. Vibration profile.

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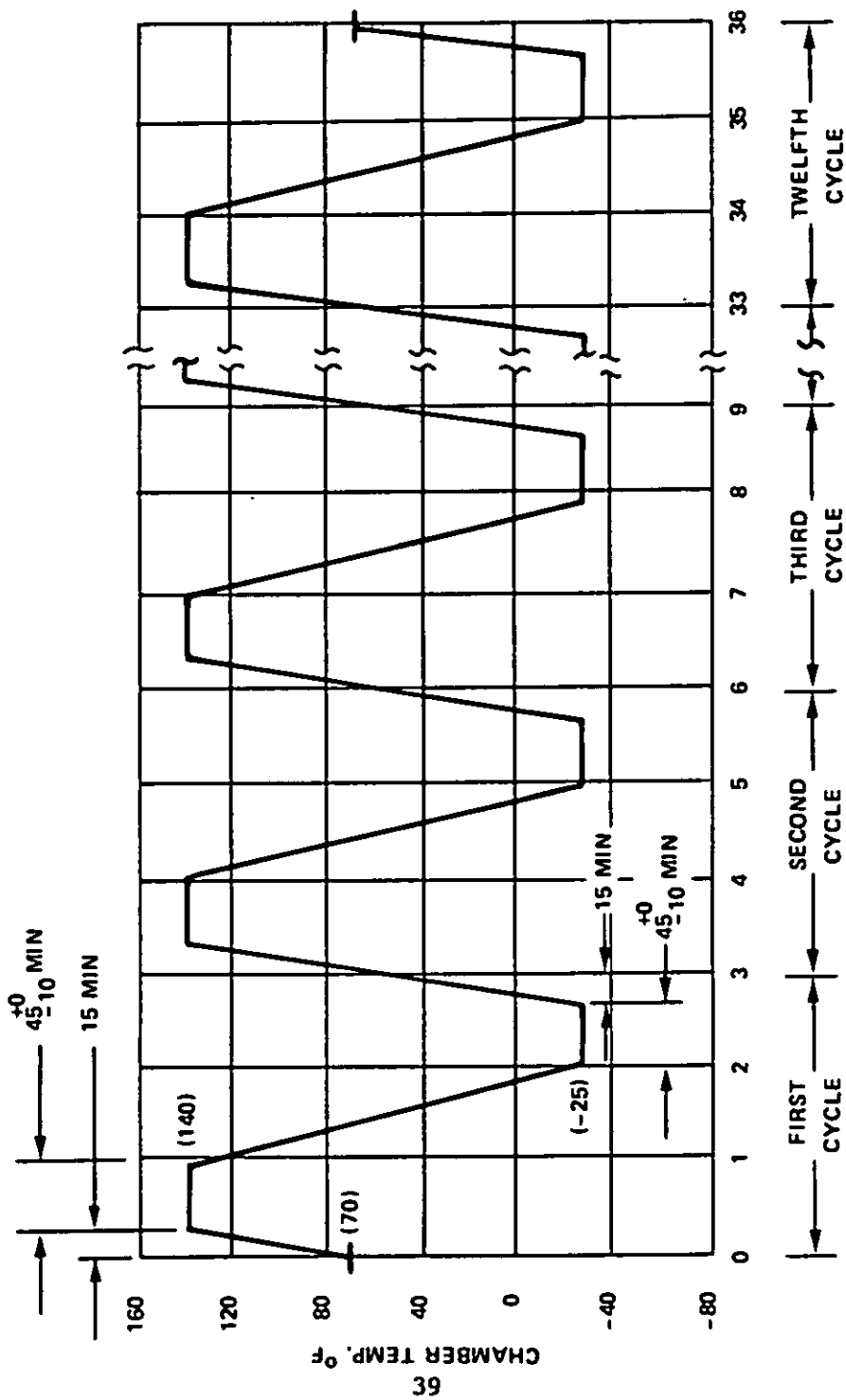


FIGURE 11. Temperature cycling.

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3.5 Environmental.

3.5.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity unit (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.5.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 10. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.5.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +125°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.5.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5).

3.5.5 Humidity. The assembly shall operate as specified herein at up to 100 percent relative humidity. (See 4.6.3.6)

3.5.6 Altitude. The assembly shall operate as specified herein under the following conditions:

a. during exposure to an atmospheric pressure equivalent to 10000 feet above sea level.

b. after exposure to an atmospheric pressure equivalent to 40000 feet above sea level for not less than 1 hour. (See 4.6.3.7)

3.5.7 Dust. The assembly shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1750 feet per minute. (See 4.6.3.8)

3.5.8 Salt fog. The assembly shall operate as specified herein after exposure to a 5 percent sodium chloride atomized spray for not less than 48 hours. (See 4.6.3.9)

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilties suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be two assemblies selected at random by the Government representative from the first five production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in table VIII.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certification of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE VIII. First article tests.

Item	Characteristic	Requirement	Test Procedure
1	Ballistic Computation	3.3.1	4.6.2.1
2	Lamp Logic Test of Ammo Select	3.3.2	4.6.2.2
3	Moving Tank Lamp Logic	3.3.3	4.6.2.3
4	Servo Drive Control	3.3.4	4.6.2.4
5	Built-in Test	3.3.5	4.6.2.5
6	Output Power and Reference Signals	3.3.6	4.6.2.6
7	ST BAR output	3.3.7	4.6.2.7
8	Fabrication	3.1	Applicable drawings - Visual
9	General Specification	3.1.2	MIL-F-13926 - Visual
10	Sealing A and B	3.4	4.6.4
11	High Temperature	3.5.3 and 3.5.4	4.6.3.3
12	Low Temperature	3.5.3 and 3.5.4	4.6.3.4
13	Vibration A	3.5.2	4.6.3.2
14	Humidity	3.5.5	4.6.3.6
15	Altitude	3.5.6	4.6.3.7
16	Shock	3.5.1	4.6.3.1
17	Dust	3.5.7	4.6.3.8
18	Salt fog	3.5.8	4.6.3.9
19	Temperature cycling	3.5.3 and 3.5.4	4.6.3.5

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4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table X.

4.5.2.2 Environmental sample and tests. Two assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table XI shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table X and only the test of item 307 in table XI shall be applied.

4.5.3.1. Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table IX and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

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4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732400.

TABLE IX. Performance tests.

Item	Characteristic	Requirement	Test Procedures
101	Ballistic Computation	3.3.1	4.6.2.1
102	Lamp Logic Test of Ammo Select	3.3.2	4.6.2.2
103	Moving Tank Lamp Logic	3.3.3	4.6.2.3
104	Servo Drive Control	3.3.4	4.6.2.4
105	Built-in Test	3.3.5	4.6.2.5
106	Output Power and Reference Signals	3.3.6	4.6.2.6
107	ST BAR Output	3.3.7	4.6.2.7
108	Sealing A	3.4	4.6.4

TABLE X. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

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TABLE XI. Environmental tests.

Item	Characteristic	Requirement	Test Procedures
303	Shock	3.5.1	4.6.3.1
304	Vibration B	3.5.2	4.6.3.2
305	High Temperature	3.5.3 and 3.5.4	4.6.3.3
306	Low Temperature	3.5.3 and 3.5.4	4.6.3.4
307	Temperature cycling	3.5.3 and 3.5.4	4.6.3.5
308	Sealing B	3.4	4.6.4

TABLE XII. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit Test Set
2	11750441	Adapter Computer Unit
3	11821748	Test Set - Immersion Leak, Purge and Fill
4	11821520	Automatic Test Assembly- Central Processor Unit
5	11821208	Fixture Chamber - Subsystem
6	11821457	Fixture Shock, Vertical and Horizontal
7	11821458	Shock Adapter, Plate Vib Horiz-Vert Plane Shock-Z
8	11823177	Unit Fixture-Computer, M60A3 Vib/Shock
9	11821471	Adapter - Handling Fixture, Altitude

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4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in table XII shall be used where specified to perform the required tests.

4.6.2 Performance tests.

4.6.2.1 Ballistic computation. Use items 1, 2 and 4 of table XII to determine compliance with 3.3.1.

4.6.2.2 Lamp logic test of ammo select. Use items 1, 2 and 4 of table XII to determine compliance with 3.3.2.

4.6.2.3 Moving tank lamp logic. Use items 1, 2 and 4 of table XII to determine compliance with 3.3.3.

4.6.2.4 Servo drive control. Use items 1, 2 and 4 of table XII to determine compliance with 3.3.4.

4.6.2.5 Built-in test. Use items 1, 2 and 4 of table XII to determine compliance with 3.3.5.

4.6.2.6 Output power and reference signals. Use items 1, 2 and 4 of table XII to determine compliance with 3.3.6.

4.6.2.7 ST BAR output. Use items 1, 2 and 4 of table XII to determine compliance with 3.3.7.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 6, 7 and 8 of table XII and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure I figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ± 4 g with a time duration of 18 ± 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

4.6.3.1.2 High intensity. Use items 7 and 8 of table XII and subject the assembly to the shock test specified in MIL-STD-810, method 516.2, procedure IV figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ± 10.0 g with a time duration of 1.5 ± 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

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4.6.3.2 Vibration. Application of Vibration A and Vibration B shall be as determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

4.6.3.2.1 Vibration A. Use items 7 and 8 of table XII and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2, procedure VIII, except that the test level shall be the vibration curve shown in figure 10 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 5 of table XII and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 125°F. The performance tests of 4.6.2 shall be applied.

4.6.3.4 Low temperature. Use item 5 of table XII and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of 4.6.2 shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 11. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

4.6.3.6 Humidity. Place the assembly in a humidity chamber and subject it to the humidity test specified in MIL-STD-810, method 507.1, procedure IV. During steps 4, 6 and 8 of procedure IV, subject the assembly to the performance tests of 4.6.2. After exposure, subject the assembly to the performance tests of 4.6.2.

4.6.3.7 Altitude. Use items 5 and 9 of table XII and subject the assembly to the altitude test specified in MIL-STD-810, method 500.1, procedure I, except that:

a. First, the assembly in a non-operating condition, shall be exposed to an atmospheric pressure equivalent to 40000 feet above sea level for a period of not less than 1 hour.

b. Second, with the chamber pressure increased to an atmospheric equivalent to 10000 feet above sea level, subject the assembly to the performance tests of 4.6.2.

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4.6.3.8 Dust. Subject the assembly to the dust test specified in MIL-STD-810, method 510.1, procedure I. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

4.6.3.9 Salt fog. Subject the assembly to the corrosion test specified in MIL-STD-810, method 509.1, procedure I. At the conclusion of this test, subject the assembly to the performance tests of 4.6.2.

4.6.4 Sealing. Pressurize the assembly with nitrogen to 5.0 ± 0.1 psig and perform either Sealing A or Sealing B leakage tests. After determination of compliance with the sealing requirement, purge the assembly with dry nitrogen at 1 psig for not less than 5 minutes and lower the pressure to 0.0 + 0.2 - 0.0 psig. Nitrogen shall be BB-N-411, type I, class 1, grade B with a dewpoint not greater than -25°F.

4.6.4.1 Test selection. Application of Sealing A and Sealing B shall be as determined by the procurement conditions of 4.5.1 and the first article conditions of 4.4.

4.6.4.2 Sealing A. Use item 3 of table XII and subject the assembly to the leakage (immersion) test specified in MIL-STD-810, method 512.1, procedure IV. Immersion time shall be not less than 5 minutes.

4.6.4.3 Sealing B. Use item 3 of table XII, place the assembly in the following environment for not less than 2 hours and verify that the pressure drop is not greater than 0.2 psig.

a.	Temperature	73° ± 2.5°F
b.	Relative humidity	50 percent ± 5 percent
c.	Atmospheric pressure	28.5 + 2.0 - 3.0 in. Hg.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732400. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The assembly is a part of the M21 Ballistic Computer.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required, the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48456B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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